

A Temporal Analysis of Cassini/VIMS Limb Brightness Profiles of Titan from 2004-2017

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ABSTRACT

We analyze 29 low-phase-angle *Cassini* Visual and Infrared Mapping Spectrometer (VIMS) full-disk observations acquired from 2004 to 2017. We fit north- and south-hemisphere radial brightness profiles with a quadratic limb-darkening law and use $u_1 + u_2$ as the signed measure of limb darkening or brightening. The north-minus-south median changes sign between northern winter and northern spring in both VIMS channels. For the visible channel the medians are 0.168 and -0.110 , and for the infrared channel they are 0.035 and -0.376 , respectively. Whole-observation bootstrap intervals exclude zero for both visible phases and for infrared spring, but not for infrared winter. The sign change remains under tested emission-angle and fit-quality rules. These results show a seasonal change in Titan’s hemispheric limb profiles while leaving detailed atmospheric retrieval to future work.

Keywords: Titan, Upper atmosphere, Seasonal phenomena

1. INTRODUCTION

Titan’s thick nitrogen atmosphere has proven to be strongly dynamic, responding to seasonal insolation trends in a complex manner. In addition to clouds (Rodríguez et al., 2009; Schaller et al., 2009; Griffith et al., 2009; Brown et al., 2010; Rodríguez et al., 2011a,b; Turtle et al., 2018) and dust storms (Rodríguez et al., 2018) in the lower atmosphere, Titan’s upper atmospheric circulation varies seasonally as evidenced by variations in its haze aerosols (Rannou et al., 2002; Tomasko et al., 2005; Seignovert et al., 2021).

From the Voyager 1 encounter in 1980, Smith et al. (1981) showed a north-south hemispheric dichotomy in the observed haze brightness. Except for distant Voyager 2 imaging (Smith et al., 1982), Titan remained unresolved until the Hubble Space Telescope (HST) based observations were again able to pick up on haze variations (Caldwell et al., 1992; Lorenz et al., 1997, 1999, 2001; Young et al., 2002; Lorenz et al., 2004; Ádámkóvics et al., 2006; Karkoschka, 2016, 2022). With adaptive optics, Titan’s hemispheric dichotomy can be measured from ground-based observatories as well (Gibbard et al., 2004; Ádámkóvics & de Pater, 2017; Nichols-Fleming et al., 2021).

The arrival of Cassini into the Saturn system in 2004 provided a new opportunity to monitor Titan repeatedly and from close range. The Imaging Science Subsystem (ISS) instrument, a fine-resolution visible-light camera, observed extensive variations of the haze with latitude and time (Porco et al., 2005; de Kok et al., 2010; Snell & Banfield, 2024). Cassini’s Visual and Infrared Mapping Spectrometer (VIMS), an imaging spectrometer, mapped haze distributions across a factor of 10 in wavelength from $0.4 - 5.2 \mu\text{m}$. VIMS also identified seasonal variations (Seignovert et al.,

2021; Vashist et al., 2023) along with diagnostic spatial features (Kutsop et al., 2022).

The different haze appearance, including the hemispheric dichotomy, results from a non-obvious interplay between haze concentration, haze single-scattering albedo, atmospheric absorption, and the vertical structure of the variation of those parameters. For instance, while the north-south dichotomy remains, the brightness ratio between the hemispheres can be different when viewed at different wavelengths — in the visible versus in the near-infrared for instance (Vashist et al., 2023). While the evidence of the vertical variation in haze and atmosphere may be represented in Titan spectra (Rannou et al., 2010), retrievals of atmospheric profiles from spectra alone remain non-unique.

Cassini employed a few different techniques that isolate the haze’s vertical structure. In particular, ISS was able to monitor conditions of the “detached haze” — a thin, discrete high-altitude layer of haze enhancement first seen by Voyager 1 (Smith et al., 1981). Lavvas et al. (2009) and Seignovert et al. (2017) constrained properties of the aerosol particles within a particular detached haze layer. The haze dropped sharply in altitude from ~ 500 km down to ~ 370 km at the time of Titan’s northern spring equinox in 2008 (West et al., 2011). The detached haze later disappeared in 2012 and reappeared later in 2016 West et al. (2018).

The other technique that reveals vertical structure uses occultations. When a star, or the Sun, passes behind the limb of Titan as viewed from Cassini, the spectral lightcurve of the occultation depends on vertical atmospheric variations. VIMS observed a set of 14 relevant occultations (Sotin et al., 2016; Robinson et al., 2014), from which atmospheric absorptions (Bellucci et al., 2009) and haze abundance could be ascertained at those particular times and places (Kim & Courtin, 2013; Maltagliati et al., 2015; Rannou et al., 2022). Observations of stellar occultations in ultraviolet wavelengths (Koskinen et al., 2011) showed in-

tricate structure unable to be adequately fit by simple haze parameterizations (Tribbett et al., 2021).

We suggest that it may be possible to use limb darkening across Titan’s spatially resolved disk at wavelengths with differing degrees of gaseous absorption to probe haze properties as a function of altitude. In this paper, we report observed limb darkening or limb brightening parameters for Titan’s haze throughout the Cassini mission, leaving detailed analysis and modeling for future efforts. We discuss our observations from the VIMS instrument in Section 2, and the methods that we use to analyze those observations as well as a simple two-layer model for their interpretation in Section 3. We then discuss the results of the analysis in Section 4 before concluding in Section 5.

2. OBSERVATIONS

As shown in Table 1, we analyze Cassini VIMS-V and VIMS-IR observations from 24 targeted flybys and 5 non-targeted flybys, spanning the Cassini probe’s 13 year operational period from 2004-2017. The VIMS instruments have been well characterized Brown et al. (2004), with the VIMS-V (Visual) channel of the using a two-dimensional array detector covering a spectral range from 0.30 to 1.05 μm with 96 distinct bands and the VIMS-IR (Infrared) channel using a one-dimensional detector covering a broader spectral range of 0.3 to 5.1 μm with 256 wavelength bands.

2.1. Cube Selection

We selected flybys using the following criteria: (1) simultaneous observations from both the VIMS-V and VIMS-IR instruments; (2) a low phase angle ($\leq 30^\circ$ preferred); (3) full-disk coverage with high limb visibility; (4) sufficient spatial resolution (200km/pixel); (5) sub-spacecraft longitude between 20°S and 20°N ; and (6) sufficient time cadence so as to obtain a comprehensive distribution of measurements during the *Cassini* mission. It is important to note that during the latter half of the *Cassini* mission, the spacecraft was in a polar orbit, which limited the number of flybys that met the above criteria, leading to the selection of non-targeted flybys, expansion of criteria, and sparser temporal coverage.

3. METHODS

3.1. SRTC++

We use 1,889 SRTC++ image cases to test how a two-layer atmosphere can change the fitted limb profile. The grid varies haze abundance, single-scattering albedo, and gas contribution in lower and upper layers. We treat the SRTC++ renderer as the source of the synthetic images and do not infer its internal physical assumptions from the image set. We rebuild $u_1 + u_2$ and the image-quality flag from every TIFF. All 1,889 quality flags match the saved run, and the median absolute coefficient drift is 4.7×10^{-10} . As a check on the mapping from inputs to limb behavior, we compare a random forest, histogram gradient boosting, and a scaled radial-basis support-vector regressor using one fixed 80/20 split. The best held-out result is from the random forest, with $R^2 = 0.990$. Its largest feature importance is the upper-layer single-scattering albedo. These model scores describe interpolation within this grid and do not validate the atmospheric assumptions of SRTC++.

3.2. Image Processing

We remove VIMS-V column striping by subtracting each column’s mean sky signal and restoring the mean background level. We do not destripe VIMS-IR or SRTC++ images. A fixed one-based band list excludes known surface windows and the long-wavelength range where fit coverage and quality decline. Eight of 352 bands fail our separate band-health check; all lie between 3.84 and 4.78 μm and are already outside the main analysis range.

We estimate the disk center from the lowest emission-angle pixels and find image north from the latitude field. We draw radial lines on a grid enlarged by a factor of five, then map the selected pixels back to the native cube. For each band we select transects 30° north and south of the projected equator on the illuminated side of the disk. We sort each profile by emission angle and retain points with $25^\circ < e \leq 89^\circ$. This is the rule used by the original production run; the old folder name suggesting a north-south boundary mask did not reflect the active path. The result is two profiles for each of 352 bands in each cube, or 20,416 profiles in total. The current fixed software stack defines the accepted line raster. We retain a row-level comparison with the earlier raster as an audit record.

3.3. Quadratic Limb Darkening Law Regression

To determine the magnitude of limb darkening and brightening within each transect, we fit the data using the known stellar Quadratic Limb Darkening Law (QLDL) from Kopal (1950); Brown et al. (2001).

$$\frac{I(\mu)}{I(1)} = [1 - u_1(1 - \mu) - u_2(1 - \mu)^2] \quad (1)$$

Where μ is the cosine of the angle between a line normal to the stellar surface and the line of sight of the observer leading to a scale from 1 to 0, one representing the center of the disk and 0 being the limb. $I(1)$ represents the intensity at the center of the disk. u_1 and u_2 are limb darkening coefficients. The limb darkening coefficients alone are not well constrained. So to measure the magnitude of limb darkening/brightening, we use $\frac{1}{I(1)} \frac{dI}{d\mu} \Big|_{\mu=0.5} = u_1 + u_2$.

We map emission angle to μ , interpolate each profile to 200 points with a shape-preserving cubic interpolator, and test the interpolated profile, a Gaussian-smoothed profile, and a moving-average profile. We fit $I(1)$, u_1 , and u_2 by nonlinear least squares, constrain only $I(1)$ to be non-negative, and retain the candidate with the highest R^2 against the sampled profile. The covariance matrix gives the variance of $u_1 + u_2$ as $\text{var}(u_1) + \text{var}(u_2) + 2\text{cov}(u_1, u_2)$. We add north and south variances when reporting the error of their difference.

Visually, the limb darkening is prevalent at lower, visible, wavelengths, while the limb brightening is more prominent at higher, near-infrared, wavelengths, though the wavelength where darkening and brightening transition seemingly evolves. The differences in limb darkening and brightening are logical when factoring the low albedo of haze at visible wavelengths and the increased haze-ratio at higher altitudes. Limb observations also reinforce existing knowledge of the north-south asymmetry’s visual shift, with a strong hemispheric dichotomy seen in most bands.

3.4. Quality, Seasonal, and Sensitivity Analysis

Table 1. North-South Boundary Observations

Year	Month	Cassini flyby	Sub-Spacecraft Latitude (°)	Spatial Resolution (km/pixel)	Phase Angle (°)
2004	10	TA	-15 N	108 km/pixel	13
2005	2	T3	-3 N	147 km/pixel	20
2005	10	T8	n/a	79 km/pixel	23
2005	12	T9	n/a	82 km/pixel	28
2006	2	T11	n/a	100 km/pixel	18
2007	5	T30	15 N	137 km/pixel	28
2007	5	T31	10 N	87 km/pixel	23
2007	6	T32	2 N	109 km/pixel	15
2007	6	T33	n/a	134 km/pixel	12
2007	8	T35	-3 N	127 km/pixel	27
2007	10	051TI ¹	14 N	207 km/pixel	26
2009	7	T58	-28 N	116 km/pixel	28
2009	8	T61	-7 N	142 km/pixel	14
2009	10	T62	-1 N	145 km/pixel	11
2010	4	T67	n/a	88 km/pixel	16
2011	4	T75	n/a	124 km/pixel	16
2011	6	T77	n/a	89 km/pixel	22
2011	12	T79	n/a	124 km/pixel	17
2012	1	T81	-8 N	107 km/pixel	23
2012	5	T83	-8 N	116 km/pixel	23
2012	6	T84	-14 N	119 km/pixel	28
2013	5	191TI ¹	-9 N	267 km/pixel	28
2014	4	T100	50 N	130 km/pixel	35
2015	7	T112	n/a	117 km/pixel	26
2015	11	T114	-1 N	99 km/pixel	26
2016	1	T115	-1 N	120 km/pixel	27
2016	12	255TI ¹	46 N	326 km/pixel	20
2017	5	273TI ¹	38 N	243 km/pixel	16
2017	6	278TI ¹	53 N	179 km/pixel	28

¹ non-targeted flyby of Titan

Fits fail the main quality gate if fitting fails, fewer than six profile points remain, a required value is not finite, or the covariance matrix is invalid. Fits with negative R^2 remain in the data as review cases but do not enter the main analysis. We apply no hard coefficient bound. Of 20,416 fits, 19,791 are eligible, 397 require review, and 228 fail. We calculate north-minus-south values only for bands where both fits are eligible.

We reduce each cube and VIMS channel to the median across allowed bands before comparing seasons, so bands do not act as independent observations. Fixed-seed bootstrap intervals resample whole cubes 5,000 times. Northern winter contains 12 cubes, northern spring 16, and northern summer one. We do not report an interval for the one-cube summer group. Our sensitivity grid repeats the full fit and analysis for emission cutoffs of 20°, 25°, and 30°, minimum R^2 values of 0, 0.25, and 0.5, and coefficient bounds of none and 10.

4. RESULTS

The main run yields 4,067 paired band measurements. In northern winter, the median north-minus-south $u_1 + u_2$ difference is 0.168 for VIMS-V (95% bootstrap interval 0.133 to 0.243) and 0.035 for VIMS-IR (-0.141 to 0.392). In northern spring, the values are -0.110 for VIMS-V (-0.331 to -0.058) and -0.376 for VIMS-IR (-0.509 to -0.265). Northern summer has only one observation and remains descriptive.

The sign change is stable across all 18 sensitivity settings. Winter medians range from 0.167 to 0.192 in VIMS-V and 0.030 to 0.072 in VIMS-IR. Spring medians range from -0.122 to -0.110 in VIMS-V and -0.377 to -0.373 in VIMS-IR. Depending on the rule, 15,380 to 19,893 fits and 3,712 to 4,070 north-south pairs remain. Two series have more than one darkening-to-brightening crossing in the least strict settings. We show all crossings but exclude multi-crossing series from scalar transition summaries.

5. CONCLUSION

Titan's north-south limb-profile difference changes sign from northern winter to northern spring in both VIMS channels. This result survives the tested profile and fit-quality choices. The visible winter and spring intervals and the infrared spring interval exclude zero; the infrared winter interval does not. The single northern-summer cube cannot support a phase-level claim. Fit covariance, whole-observation resampling, band-health checks, and the full sensitivity grid now make the limits of these results ex-

plicit. A later physical retrieval should test whether changes in upper-layer haze scattering can reproduce the observed time and wavelength pattern.

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